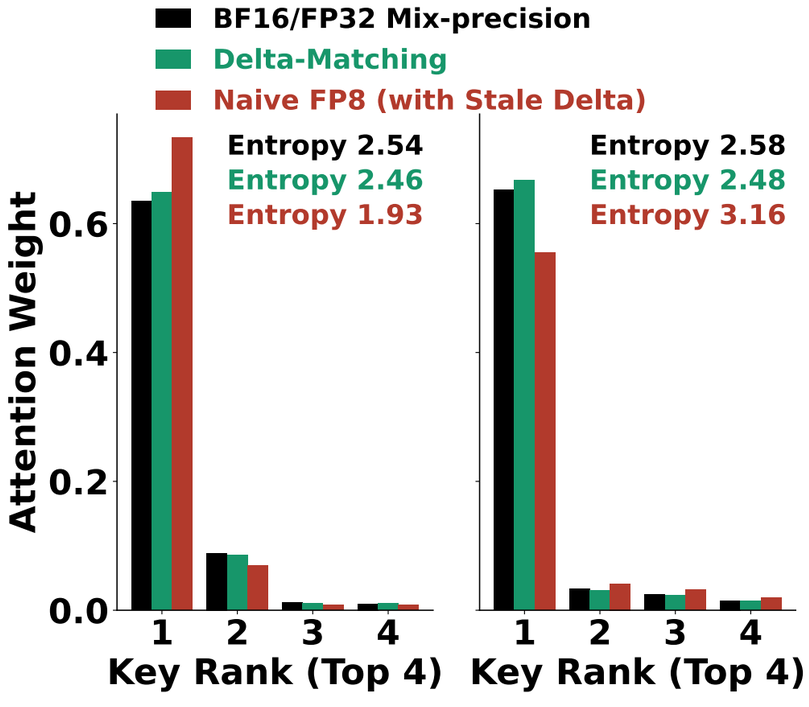
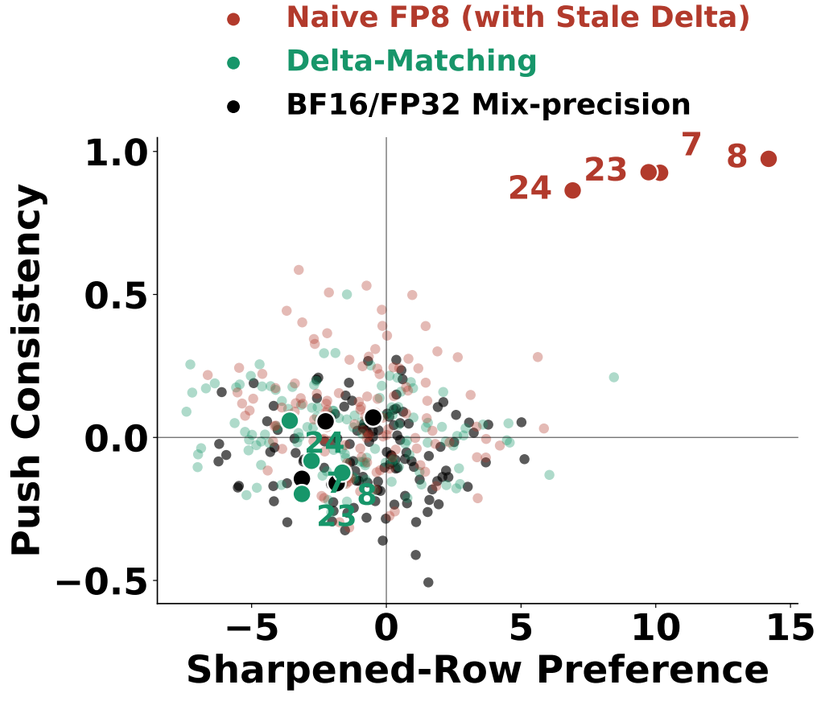
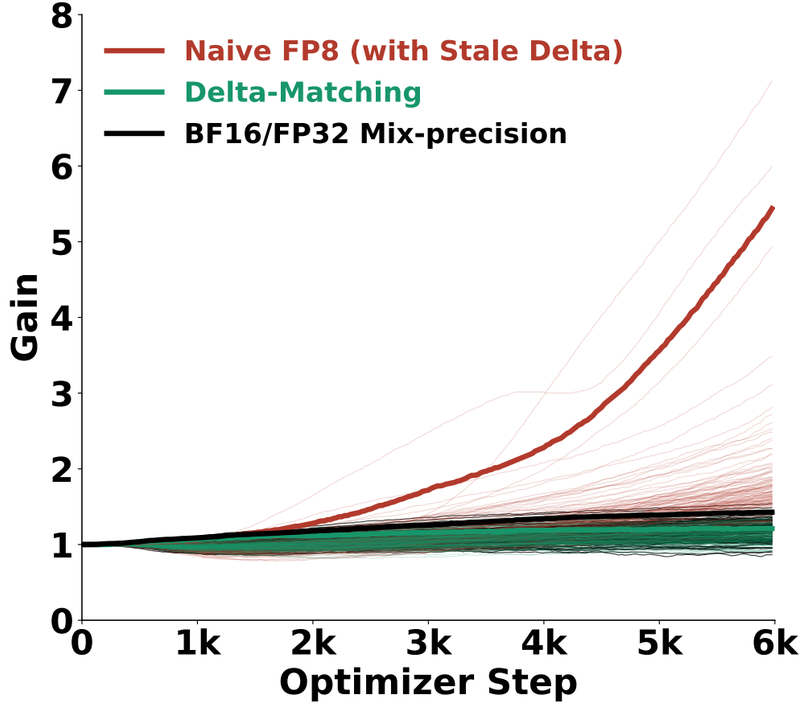

BF16/FP32 Mix-precision
Delta-Matching
Naive FP8 (with Stale Delta)
Entropy 2.54
Entropy 2.46
Entropy 1.93
Entropy 2.58
Entropy 2.48
Entropy 3.16
Attention Weight
0.0
0.2
0.4
0.6
1
2
3
4
Key Rank (Top 4)
Key Rank (Top 4)

(a)


Naive FP8 (with Stale Delta)
Delta-Matching
BF16/FP32 Mix-precision
Push Consistency
1.0
0.5
0.0
−0.5
24
23
7
8
−5
0
5
10
15
Sharpened-Row Preference

(b)


Naive FP8 (with Stale Delta)
Delta-Matching
BF16/FP32 Mix-precision
Gain
0
1
2
3
4
5
6
7
8
0
1k
2k
3k
4k
5k
6k
Optimizer Step

(c)